

The influence of the grazing flow and sound incidence direction on the acoustic characteristics of double Helmholtz resonators

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ABSTRACT

Restricted by the principle of resonance, Helmholtz resonators can effectively absorb noise only near the resonance frequency. It is an effective way that the coupling design of the multiple resonators to broaden the absorption band. In practical application, Helmholtz resonators are usually arranged in the flow duct, so that will be affected by the working condition. In order to investigate the influence of the grazing flow and the sound incident direction on the acoustic performance, both numerical simulations and experiments are conducted. A three-dimensional model of the double Helmholtz resonators is established. By solving the linearized Navier–Stokes (LNS) equations, the sound transmission loss of the system in two sound incident directions is predicted where the mean flow Mach number is 0 and 0.1, respectively. The absorption, reflection, and transmission coefficients are further obtained. Based on the comparison of the flow fields and sound fields, the mechanism of the variation of the acoustic characteristics is analyzed. The experimental results show that the maximum sound transmission loss decreases by 16.9 dB, and the resonance frequency increases by 59 Hz. Both numerical and experimental results indicate that the grazing flow weakens the ability of noise suppression and the low-frequency characteristics of the double Helmholtz resonators in this paper. The inherent asymmetric absorption/reflection phenomenon vanishes, and the acoustic characteristics of the coupled system are dominated by the background flow. While the exchange of the sound incidence direction can mitigate the penalty of the grazing flow on the coupled system.

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1. Introduction

Noise control is an important research endeavor in the field of acoustics. Generally, the sound absorber is mainly divided into two categories: the porous materials and the resonators. As an effective sound absorption structure, the Helmholtz resonator (HR) has been widely used in the fields of automobiles and aviation [1–5]. Restricted by the principle of resonance, the HR can only achieve perfect sound absorption near the resonance frequency. Recently, the concept of acoustic metamaterials has been widely developed due to their extraordinary sound absorption characteristics, and the researchers are devoted to breaking through the performance bottleneck of existing sound absorbers [6–12]. In order to broaden the sound absorption frequency band, many HR variants have emerged [13,14]. Selamet et al. [15] investigated the acoustic performance of the HR with an extended neck, it was

found that the resonance frequency can be modified by the geometric parameter of the extended neck with the unchanged cavity volume. Zhou et al. [16] set a flexible plate in a resonance cavity, the coupled system enhances the sound absorption and dissipation in the specific frequency, and broadens the bandwidth of the resonance frequency. Huang et al. [17] employed internal intubations and combined a series of crimping cavities, the resulting compact system has the characteristics of small size and a wide sound absorption band. Long et al. [18] studied the asymmetric sound absorption characteristics of double HRs and found that under the incident sound waves in different directions, the coupling system can present the asymmetric sound absorption/reflection.

It is worth noting that the HRs are commonly placed on the sidewall of pipe systems or duct networks in practical applications. The duct is usually accompanied by airflow. It is inevitable that the flow would affect the acoustic performance of the HRs, which attracts the researcher's interest [19,20]. Selamet et al. [21] studied the influence of the grazing flow on the sound absorption performance of a single HR, both numerical and experimental results

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show the grazing flow decreases the peak sound transmission loss and shifts the resonance frequency to a higher frequency value. Holmberg et al. [22,23] analyzed the acoustics of the inlet section of a side-branch mounted resonator and observed the whistle phenomenon since the vortex-sound interaction amplified incoming sound waves. Zhao et al. [24] investigated the acoustic damping effect of the perforated liners with the bias-grazing flow. The experimental results presented that grazing flow reduced the maximum acoustic power absorption, while the bias flow could increase the damping effect. Du et al. [25] studied the influence of the airflow Mach number on the acoustic performance of the double-ring resonator. They found that the peak value of the sound transmission loss decreases obviously with the increase of the Mach number. These researches suggest that the grazing flow will weaken the sound absorption of the HRs and even amplify the noise in some cases.

More specifically, the HRs are usually employed to tune the acoustics of the intake and exhaust system. For the intake system, the sound incidence direction is opposite to the flow direction. For the exhaust system, the sound incidence direction and the flow direction keep the same. The different directions may have the potential impact on noise suppression, which also motivates our research interest. In this work, the research object is an asymmetric sound absorber composed of two adjacent Helmholtz resonators (HRs), which can achieve asymmetric absorption/reflection and have versatile applications [18]. we focus on the influence of the grazing flow and the sound incoming direction and evaluate the acoustic characteristics of the double HRs coupled system by using numerical and experimental methods. This paper is organized in the following order. Firstly, details on the numerical model are described in Section 2. Secondly, the experiment setup and data processing methods are presented in Section 3. Thirdly, results for the acoustic and flow characteristics are discussed in Section 4. Finally, the key findings and conclusions are summarized in Section 5.

2. Numerical model

2.1. Geometric description

The present study establishes a three-dimensional model comprising a rectangular duct and two HRs in commercial software COMSOL Multiphysics 5.5. As shown in Fig. 1, the axial length

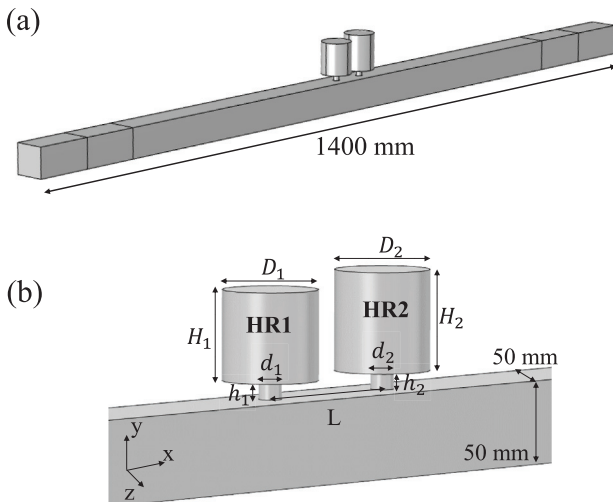


Fig. 1. Geometric configuration of the numerical models, (a) Flow duct, (b) Double Helmholtz resonators.

and the side length of the square section duct are 1400 mm and 50 mm, respectively. Double Helmholtz resonators (HR1 and HR2) are embedded at the same side of the duct and centered in the z -direction. The geometric parameters and resonant frequencies of two HRs are listed in Table 1. Note that, the only geometric difference between HR1 and HR2 is the cavity depth, which is 53.5 mm and 60 mm, respectively. The resonant frequencies of HRs are evaluated by $\omega^2 = c^2 S / V l_{\text{eff}}$ [26,27], where c is the speed of sound in air, S and l_{eff} are the neck area and effective length of the neck respectively, V is the cavity volume.

2.2. Determination of acoustic scattering coefficients

As illustrated in Fig. 2, the numerical model considers the influence of the grazing flow and sound incident direction on the acoustic characteristics of the system. The distance between the terminations of the duct and the HRs is enough long, so as to ensure the fully developed flow in the duct and the plane wave far from the HRs. The flow direction is fixed from left to right (along the positive axis), and the mean flow Mach number M is set as 0 (no flow) or 0.1. The incident direction of the sound wave includes the positive direction and the negative direction. So there are 4 cases in total. For the sake of brevity, we name them Case 1 ($M = 0$, left incident sound), Case 2 ($M = 0$, right incidence sound), Case 3 ($M = 0.1$, left incidence sound), and Case 4 ($M = 0.1$, right incidence sound) respectively. They are listed in Table 2.

The frequency of the incoming sound waves ranges from 300 Hz to 600 Hz, which is lower than the cut-off frequency of the duct (3430 Hz). Due to the HRs, the incident sound energy will be reflected and absorbed. Moreover, the introduction of the grazing flow complicates sound propagation. To definitely analyze the acoustic properties of the system, we define the reflectance (R), transmittance (T), absorptance (A), and sound transmission loss (STL),

$$R = \left| \frac{P_{r1}}{P_i} \right|^2, \quad (1)$$

$$T = \left| \frac{P_{t2}}{P_i} \right|^2, \quad (2)$$

$$A = 1 - R - T, \quad (3)$$

$$\text{STL} = 20 \lg \left| \frac{P_i}{P_{t2}} \right|. \quad (4)$$

2.3. Governing equations

The governing equations of the numerical model are linearized Navier–Stokes (LNS) equations, which in the frequency domain are defined as,

$$i\omega \rho' + \nabla \cdot (\rho \mathbf{u}_0 + \rho_0 \mathbf{u}') = \mathbf{M}, \quad (5)$$

$$\rho_0 (i\omega \mathbf{u}' + (\mathbf{u}' \cdot \nabla) \mathbf{u}_0 + (\mathbf{u}_0 \cdot \nabla) \mathbf{u}') - \nabla \cdot \boldsymbol{\sigma} = \mathbf{F}. \quad (6)$$

They are the continuity equation and momentum equation respectively. The energy equation is omitted because the perturbation of pressure and density can be regarded as an adiabatic process in this study [28]. Wherein, $\boldsymbol{\sigma} = -p' \mathbf{I} + \mu_t (\nabla \mathbf{u}' + (\nabla \mathbf{u}')^T) + (\mu_b - \frac{2}{3} \mu_t) (\nabla \cdot \mathbf{u}') \mathbf{I}$, $\mathbf{I}$ is the identity matrix, μ_t is the total viscosity and μ_b is the bulk viscosity. ρ' , p' , $\mathbf{u}'$ are the fluctuating terms of density, pressure, and velocity. Where, $p' = c_0^2 \rho'$, c_0 is the adiabatic speed of sound. Items with subscript 0 (eg. $\mathbf{u}_0$) represent background flow parameters, which are obtained by solving the RANS

Table 1

Geometric parameters and resonant frequencies of the Helmholtz resonators.

	Cavity height H_i (mm)	Cavity diameter D_i (mm)	Neck height h_i (mm)	Neck diameter d_i (mm)	Axial Spacing L (mm)	Resonant frequency f_i (Hz)
HR1	53.5	50	11	12	56	391
HR2	60	50	11	12	56	376

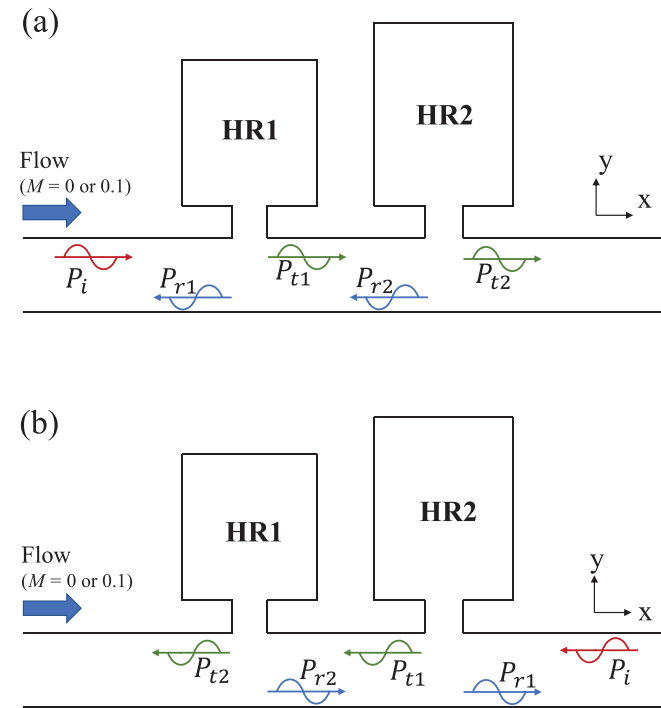


Fig. 2. Schematic of the numerical models. P_i : incidence, P_r : reflection, P_t : transmission. (a) Same direction for the flow and the incoming sound wave (Case 1 & Case 3), (b) different directions for the flow and the incoming sound waves (Case 2 & Case 4).

Table 2

Directional characteristics of each case.

Case	Flow direction	Sound direction
1	n/a	→
2	n/a	←
3	→	→
4	→	←

based on the SST $k-\omega$ model. The terms on the right side of the equations (**M** and **F**) denote the source terms. In this study, the background temperature $T_0 = 293.15$ K, the initial density $\rho_0 = 1.205$ kg/m³, reference pressure $p_{ref} = 1$ atm, air viscosity $\mu_0 = 1.81 \times 10^{-5}$ kg/(m · s).

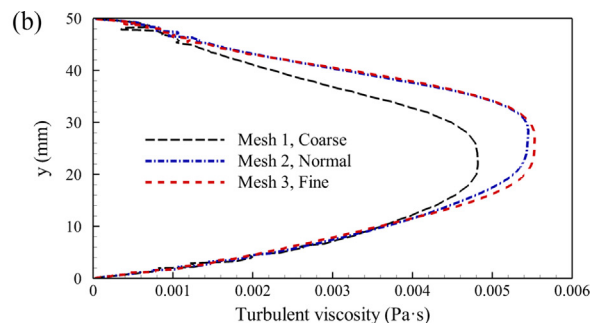
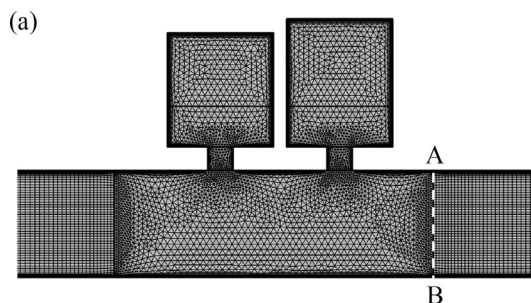


Fig. 3. (a) CFD mesh topology (Mesh 2, Normal), (b) Numerical results of turbulent dynamic viscosity for CFD mesh independence.

2.4. Mesh independence

The mesh independence studies are conducted to ensure the correctness of the numerical results. In this paper, the flow field data are first solved, and the velocity, pressure, and vortex viscosity of the flow field are imported into the LNS equations to solve the sound field data. Therefore, the computational mesh includes the CFD mesh and acoustic mesh. Fig. 3(a) shows a CFD mesh topology (labeled Mesh B, Normal), where unstructured grids are employed in the HRs and local main flow duct to accommodate complex geometric boundaries, and other domains are divided into square structure grids. Local mesh refinement is performed in the necks and all boundary layers. The mesh with different total numbers can be obtained by reducing the element size. Three CFD meshes are constructed in this paper, and the specific parameters are listed in Table 3. Fig. 3(b) shows the numerical results of turbulent dynamic viscosity obtained based on different meshes, where $y = 0$ and $y = 30$ correspond to points B and A in Fig. 3(a). As shown in Fig. 3(b), the curves of Mesh 2 case and Mesh 3 case are almost identical, with a slight difference in the middle of the main pipe. Considering the calculation time cost comprehensively, Mesh 2 is used to calculate flow field data in this paper.

Similarly, mesh independence of sound field computing is also studied. The numbers of the three meshes are listed in Table 4. Fig. 4(a) shows the topology of the acoustic mesh (labeled Mesh2, Normal). Compared with the CFD mesh, the number of boundary layer meshes is reduced, and the mesh is further refined in the neck and nearby area to capture the change of acoustic variables. Fig. 4(b) shows the calculation results when the sound wave is incident from the left side. The curves of the Mesh B case and Mesh C case are completely consistent, indicating that Mesh B can meet the calculation requirements. Therefore, Mesh B is adopted in the calculation of the acoustic data in this paper.

3. Experimental measurement

3.1. Experimental setup

As shown in Fig. 5, the test facility involves a flow duct with a square cross-section. The side length is 50 mm, corresponding to that in the aforementioned numerical model. The gauges are installed on the side of the flow duct and form mirror distribution around the HRs. Two anechoic terminations distribute at the front

Table 3

CFD mesh parameters.

	Number of elements	Minimum element size (m)	Label
Mesh 1	245011	3.5×10^{-4}	Coarse
Mesh 2	800305	1.5×10^{-4}	Normal
Mesh 3	1143220	2×10^{-5}	Fine

Table 4

Acoustic mesh parameters.

	Number of elements	Minimum element size (m)	Label
Mesh A	283669	1.5×10^{-3}	Coarse
Mesh B	635762	5×10^{-4}	Normal
Mesh C	931110	2×10^{-4}	Fine

and rear ends of the flow duct, form the non-reflective boundary condition. The airflow is already fully developed upstream of the HRs, and the mean flow Mach number is evaluated by using a pitot tube. When meeting the target flow condition, the pitot tube can be moved out of the flow duct, so as to avoid extra sound scattering. The HRs are fabricated by 3D-printing technology, see Fig. 6. The wall of HRs is enough thick that can be regarded as an acoustic hard boundary.

3.2. Measurement techniques

As mentioned before, the flow direction is fixed, and there are two pairs of loudspeakers. Only plane wave propagates inside the duct below the cut-off frequency of 3430 Hz. In this work, the incident sound pressure level (SPL) is 110 dB. By driving one pair of speakers and keeping the other silent, the incoming direction of the plane wave can be changed. When the left speakers (L1) work, the microphone A1 and A2 are employed to calculate the complex pressure reflection coefficient r of the system by means of the two-microphone method [29]. Since the existence of the mean flow, it is necessary to consider the influence of flow on wavenumber. For sinusoidal time dependence ($e^{i\omega t}$), the complex sound pressure reflection coefficient can be described as [30,31],

$$r = \frac{He^{ik^+(l+s)} - e^{ik^+l}}{e^{-ik^-l} - He^{-ik^-(l+s)}}, \quad (7)$$

where, $H = p_2/p_1$ is the transfer function between the two sound pressure signals p_1 and p_2 , which are measured by using Microphone A1 & A2 (or B1 & B2) respectively. s is the axial distance between A1 & A2 (or B1 & B2), and l is the distance between Microphone A1 (or B1) and the inlet cross-section corresponding to the coupled HRs. k^+ and k^- present the axial wavenumbers in the positive and negative directions respectively, they can be uniformly written as [32],

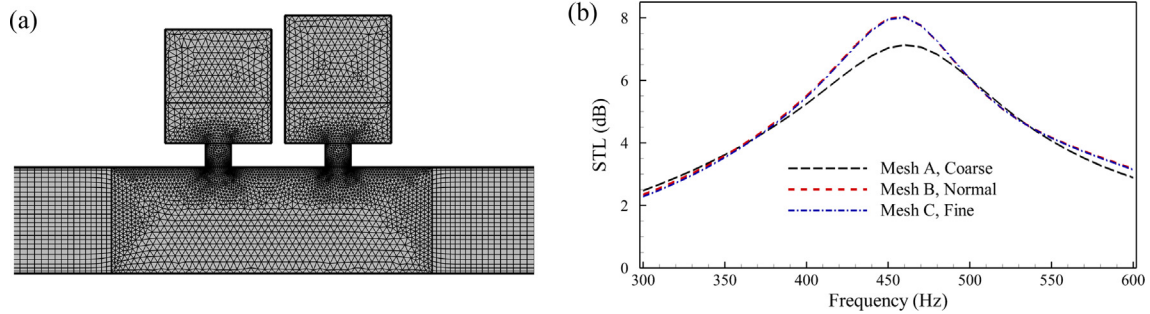


Fig. 4. (a) Acoustic mesh topology (Mesh B, Normal), (b) Numerical results of STL for acoustic mesh independence.

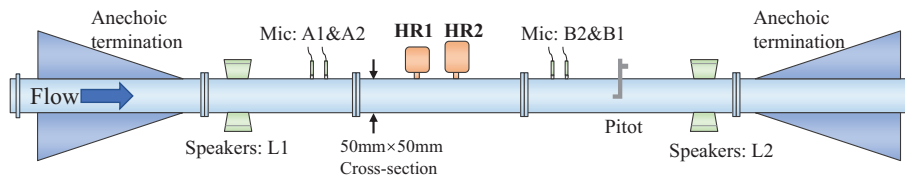


Fig. 5. Schematic of the experimental facilities.

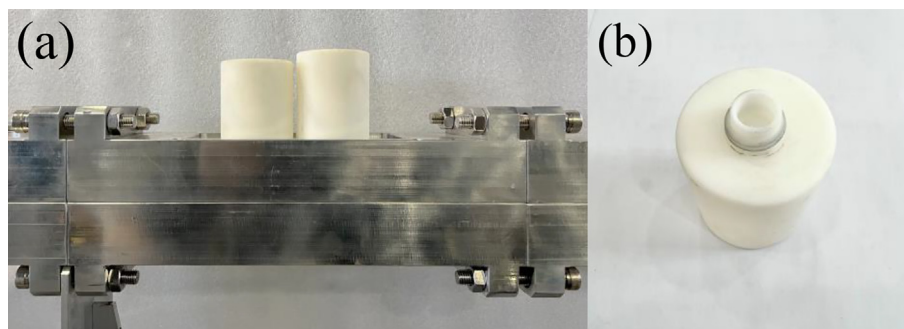


Fig. 6. Photos of the test specimen, (a) Double Helmholtz resonators fixed to the flow duct, (b) a Helmholtz resonator fabricated by using 3D-printing technology.

$$k^{\pm} = \frac{\omega/c_0 - i(\alpha + MF/2d + i\alpha)}{1 \pm M}, \quad (8)$$

where, M is the mean flow Mach number, in this study, $M = 0$ or 0.1 . $\alpha = (2\omega\mu_0/\rho_0)^{0.5}/c_0D_h$ is the thermoviscous attenuation constant. D_h is the hydraulic diameter of the flow duct. $F = 0.0072 + 0.612/\text{Re}^{0.35}$ is the Froude's friction factor, Re is the Reynolds number. Once obtaining the complex sound pressure reflection coefficient, the incoming sound pressure can be calculated by using the formula:

$$p_i = \frac{p_2}{e^{jk^+l} + re^{-jk^-l}}. \quad (9)$$

Likewise, for the downstream Microphone B1 & B2, Eq. 8 is still used to obtain the sound transmission pressure p_t . Finally, the sound transmission loss of the system is yielded by means of Eq. 4. When the right speakers (L2) work, the data processing method keeps similar, just switch the microphones of the upstream group and downstream group.

4. Results and discussions

Both numerical and experimental results are presented in this section. There are four cases including the mean flow Mach number $M = 0$ or 0.1 , and two sound incidence directions for each flow condition.

4.1. Comparison of sound transmission loss

Sound transmission loss (STL) is the most intuitive parameter to evaluate the performance of a muffler system. Fig. 7 illustrates the comparisons of STL between solutions of LNS equations and experimental results for the four cases. As seen from these curves, the tendency of STL can be predicted well. As shown in Fig. 7(a), for the no-flow condition, Case1 and Case2 have identical STL curves with two peaks. Since the intrinsic impedance of the coupled HR is not affected by the sound incident directions. The first peak is 15.1 dB and the corresponding frequency is 376 Hz, which is close to the resonance frequency of HR2 (373 Hz). The second peak is 25.5 dB and the corresponding frequency is 397 Hz, which is close to the resonance frequency of HR1 (391 Hz). It can be seen that the muffler system has good sound absorption performance in the frequency band between the two resonant frequencies through the coupling effect of the HRs. When the mean flow Mach number M equals 0.1 (Case3 and Case4, see Figs. 7(b) and (c)), the two peaks of STL merge because of the flow-induced damping, and the corresponding frequencies tend to increase. And then, to make matter worse, the peak values of STL for Case3 and Case4 decrease to 8.6 dB at 456 Hz and 10.5 dB at 434 Hz respectively. These features suggest the grazing flow weakens the sound suppression performance of the coupled system and shift the bandwidth to a higher frequency. Specifically, the maximum STL reduction and frequency shifts are 16.9 dB and 59 Hz. These tendencies are similar to the results of the single HR obtained by Selamet et al. [21]. But the different sound incidence directions show the different STL. In order to investigate the potential causes, further analysis is conducted in the next part.

4.2. Numerical results of reflectance/transmittance/absorptance

Fig. 8 shows the reflectance (R), transmittance (T), and absorptance (A) of four cases. For Case1 (see Fig. 8a), the absorptance A equals 0.9 at 376 Hz (the first resonant frequency), which means almost all the sound energy is absorbed, and the system can be seen as a quasi-perfect sound absorber. Besides, the reflectance R equals 0.87 at 397 Hz (the second resonant frequency), which sug-

gests the reflection is dominant in the system. For case 2 (see Fig. 8(b)), the absorptance A tends to zero at all computational ranges, and the reflectance R tends to 1 at the resonant frequency, the system can be seen as a quasi-perfect sound reflector. Comparing Fig. 8(a) and Fig. 8(b), the transmittance T is completely identical. The incident directions cannot change the transmissive energy downstream of the system, only decide whether the sound is reflected or absorbed at the first resonant frequency. For Case3 (see Fig. 8(c)), there occurs a peak at 456 Hz for the absorptance A curve. Compared with Case1, the maximum value of absorptance decreases obviously, and the corresponding frequency increases by 80 Hz. The transmittance curve varies slightly, ranging from 0.18 to 0.2. So, the frequency at the valley of transmittance corresponds to the peak point of absorptance. For Case4 (see Fig. 8(d)), the peak value of the absorptance curve is 0.67 at 400 Hz. Compared with Case2, the absorptance curve is no longer a near-zero curve. The flow effect induces the quasi-perfect reflection to disappear. It is noted that the peak frequencies of absorptance (400 Hz) and reflectance (460 Hz) are not coincident. The minimum transmittance

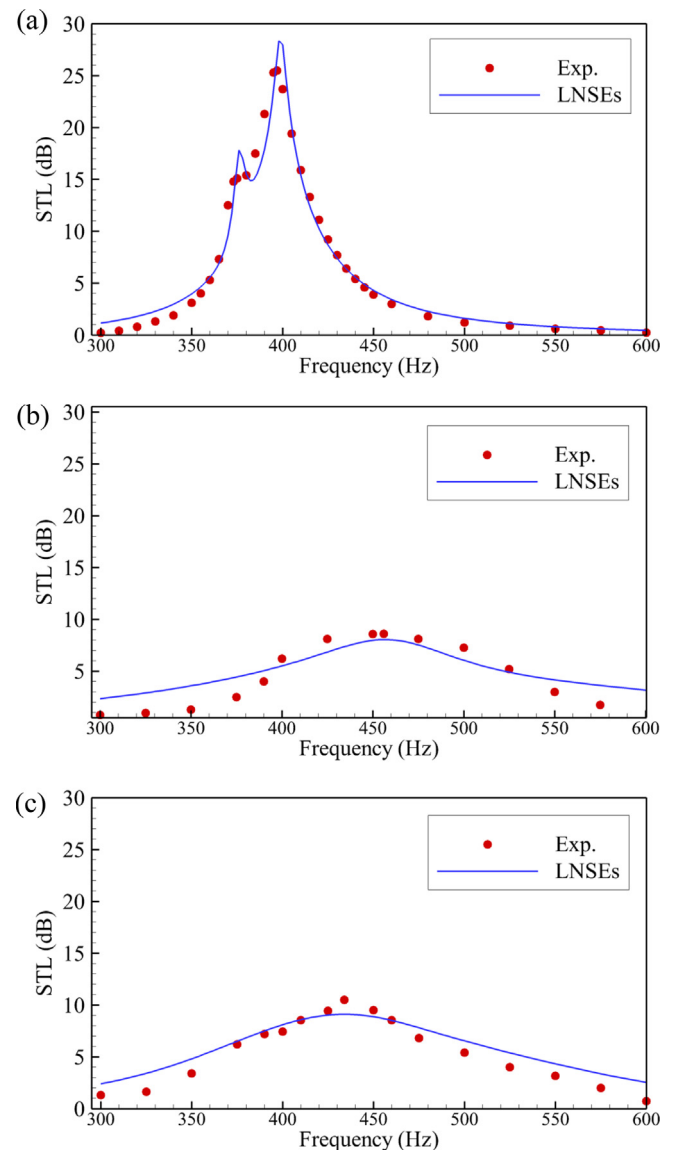


Fig. 7. Sound transmission loss of all cases, SPL = 110 dB. (a) Without flow (Case 1 & Case2), (b) with flow, $M = 0.1$, left incidence sound (Case3), (c) with flow, $M = 0.1$, right incidence sound (Case4).

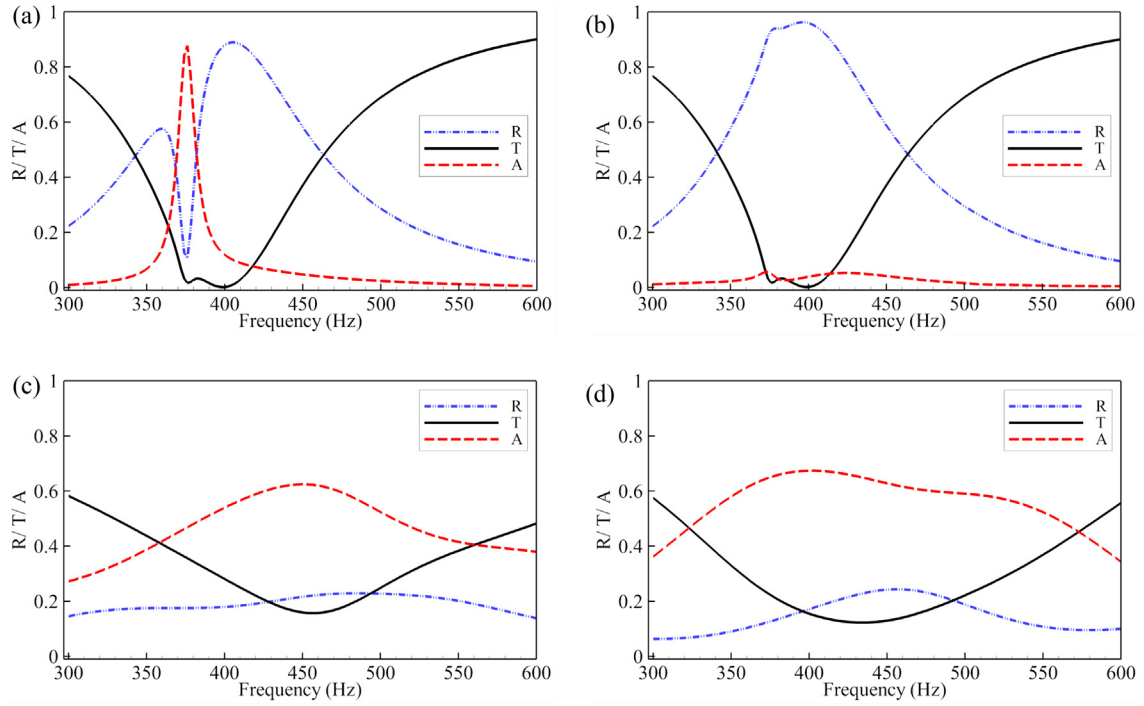


Fig. 8. Contours of the sound pressure, SPL = 110 dB. (a) Without flow, left incidence sound (Case 1), (b) without flow, right incidence sound, (Case 2), (c) with flow, $M = 0.1$, left incidence sound (Case 3), (d) with flow, $M = 0.1$, right incidence sound (Case 4).

point emerges at 434 Hz, and the transmittance curve is gentler than it is in Case 2. Therefore, when the incident sound and the air-flow go ahead in the opposite direction, the broadband characteristic of the system is slightly better than that in the same direction. In addition, based on the results of these cases, the introduction of the grazing flow dominates the acoustic performance of the system, and the asymmetric absorption/reflection features vanish.

4.3. Numerical results of flow field

In this study, the background flow field is calculated by solving the RANS based on the SST $k-\omega$ model. The flow direction is unchanged (from left to right). Fig. 9 illustrates the contours of the velocity, static pressure, and vorticity magnitude at the bisector cross-section. The thickness of the wall shear layer near

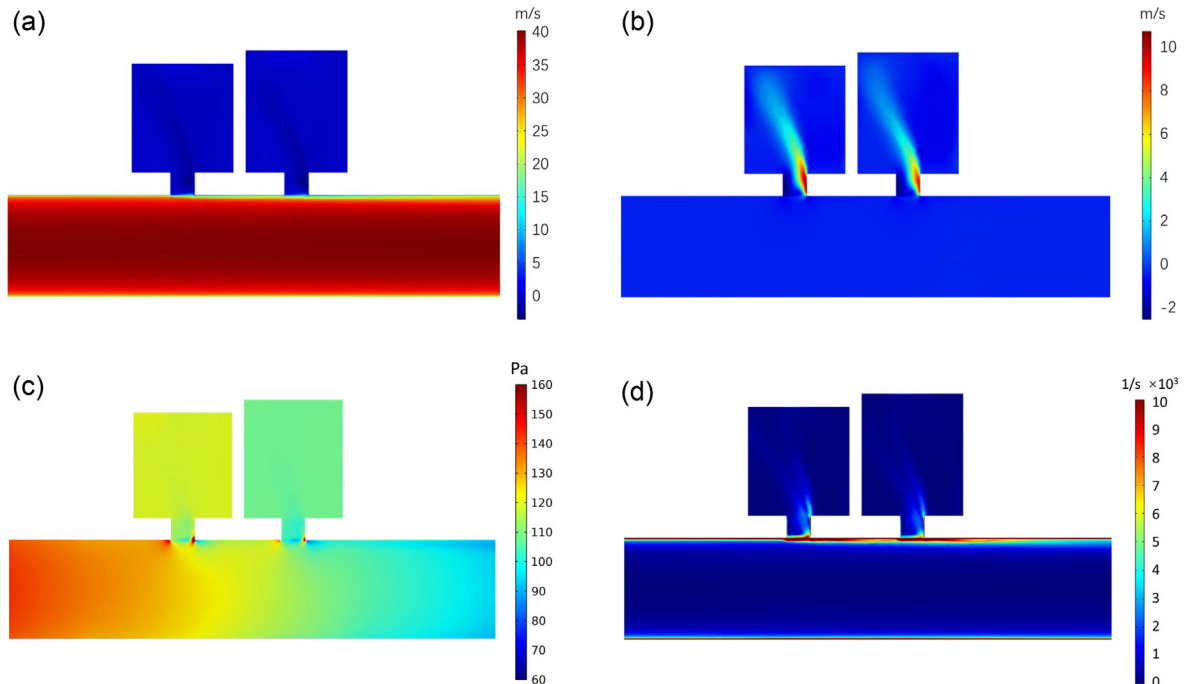


Fig. 9. Contours of the flow field, Mach number $M = 0.1$. (a) Velocity magnitude in the x-direction, (b) velocity magnitude in the y-direction, (c) static pressure, (d) vorticity magnitude.

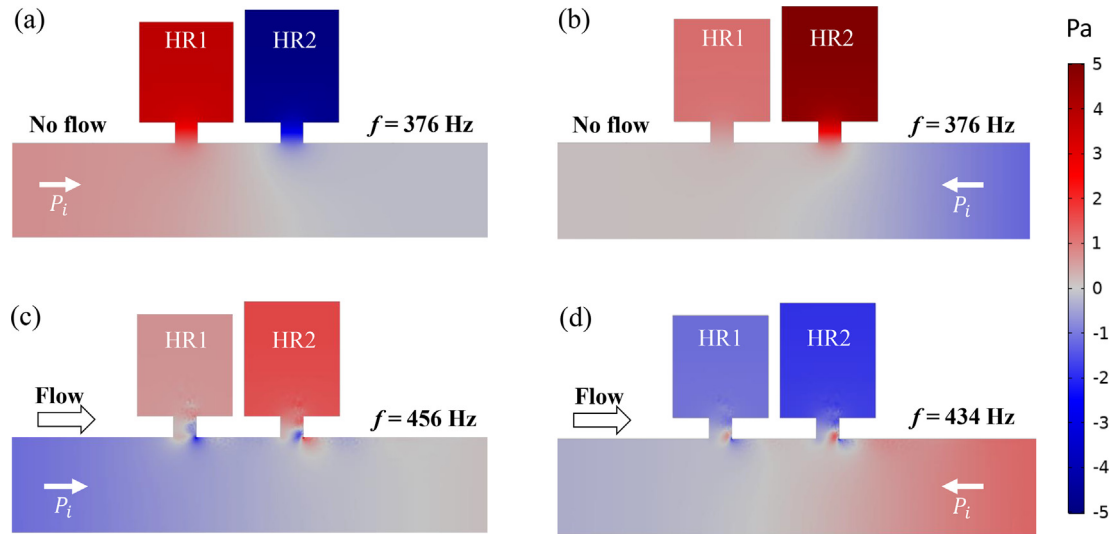


Fig. 10. Contours of the sound pressure, SPL = 110 dB. (a) Without flow, left incidence sound (Case 1), (b) without flow, right incidence sound (Case 2), (c) with flow, $M = 0.1$, left incidence sound (Case 3), (d) with flow, $M = 0.1$, right incidence sound (Case 4).

the HRs gradually increases, see Fig. 9a. Due to the flow impingement, the y-direction velocity gradients are emerged at the downstream walls of the two necks and their nearby zones (see Fig. 9(b)). It is expected that the velocity gradient of HR1 is larger than those of HR2 because of the viscous loss. Fig. 9(c) shows the static pressure distribution with the atmosphere as a reference. There is a significant local pressure rise at the connection between the necks and the main flow duct, owing to the stagnation and separation of the flow. This obvious pressure gradient will complicate the local sound wave transmission. As shown in Fig. 9(d), there is sharp vorticity variation along the upper wall of the main duct. This is due to the flow separation and the back-flow. The rapid increase of the vorticity means the background flow could strongly participate in the sound energy dissipation.

4.4. Numerical results of sound pressure

Fig. 10 shows the sound pressure contours of four cases at their resonant frequency. When $M = 0$ (no flow condition), we focus on the first resonant frequency (376 Hz) due to the existence of the asymmetric absorption/reflection phenomenon. For Case1, see Fig. 10(a), HR1 and HR2 are all resonant, and the phases of each other are opposite. Here, the sound energy is absorbed by the HRs due to viscous dissipation. For Case2, see Fig. 10(b), since the HR2 with “near-zero” impedance can be regarded as a soft boundary, almost all the incident sound waves are reflected once reaching the HR2 position. When there exists the grazing flow, the sound pressure contours obviously change. For Case3, see Fig. 10(c), it is noted that the vortex structure emerges at the neck of HR1. After hitting the sharp edge of the neck, one part of the vortex moves to the cavity and the other part continues to propagate downstream. When the vortex is transferred to HR2, the movement pattern is similar to that at HR1. The vortex propagates downstream and dissipates gradually. For Case4, see Fig. 10(d), although the sound incident direction is opposite to the flow direction, the vortex structure is still blown to the sharp edges downstream of the necks. This feature presents the grazing flow is stronger than the incident sound waves and explains the similarity of the acoustic characteristics ($R/T/A$) between Case3 and Case4.

4.5. Numerical results of sound perturbation velocities

The absorption and transition of sound energy must be accompanied by sound perturbation. The sound perturbation velocity field could show the spatial distribution of energy exchange more intuitively. Fig. 11 illustrates the contours of sound perturbation velocities at resonant frequencies and the spectrum of mean normal velocity magnitude $|v|$ at the middle cross of the necks respectively. For Case1, see Fig. 11(a), the sound perturbation is concentrated at the neck of HRs with an opposite phase, the curves below are the mean normal sound perturbation velocity at the cross-section of necks (dash lines in the Fig. 11(a)) under all frequencies. It is obvious that the peak values emerge at 376 Hz, corresponding to the quasi-perfect absorption. At this time, there exists a severe relevant movement between air particles and the wall. The coupled system transfers the sound energy into the heat by the viscous effect and dissipates eventually. For Case2, the magnitude of normal velocity is smaller than it is in Case1 at all frequencies, especially for the HR1. This feature confirms the near-zero reflectance at all frequencies. As shown in Fig. 11(c), the periodic sound perturbation velocity of Case3 is mainly distributed at the necks and nearby positions. Like the pressure and vorticity contour mentioned before, the normal velocity is blown into the right-side wall of the neck. Based on the theory of vortex sound under low Mach number, the mechanism of sound absorption is that the sound energy is transferred into the vortex and dissipates eventually by the interaction. And at 456 Hz, the velocity magnitude of HR1 is larger than it of HR2, which means HR1 is the leading role in sound absorption. For Case4 (see Fig. 11(d)), the contour under the resonant frequency (434 Hz) is similar to it of Case3. The flow direction controls the distribution of the perturbation velocity. Compared Fig. 11(c) with Fig. 11(d), the curve of HR2 is almost not changeable, but the curve of HR1 drop-down obviously under high frequency and the peak no longer exists. When there's grazing flow in the duct, the exchange of sound incident direction mainly affects the acoustic characteristics of HR1.

In addition, the acoustic perturbation velocity contour also indicates the physical mechanism of the resonant frequency shift. As mentioned above, compared Case1 with Case3, the mean flow increases the resonant frequency by 59 dB. In the absence of flow (see Fig. 11(a)), the fluid in the neck of each HR extends into the main flow duct, suggesting that the effective length of the neck is

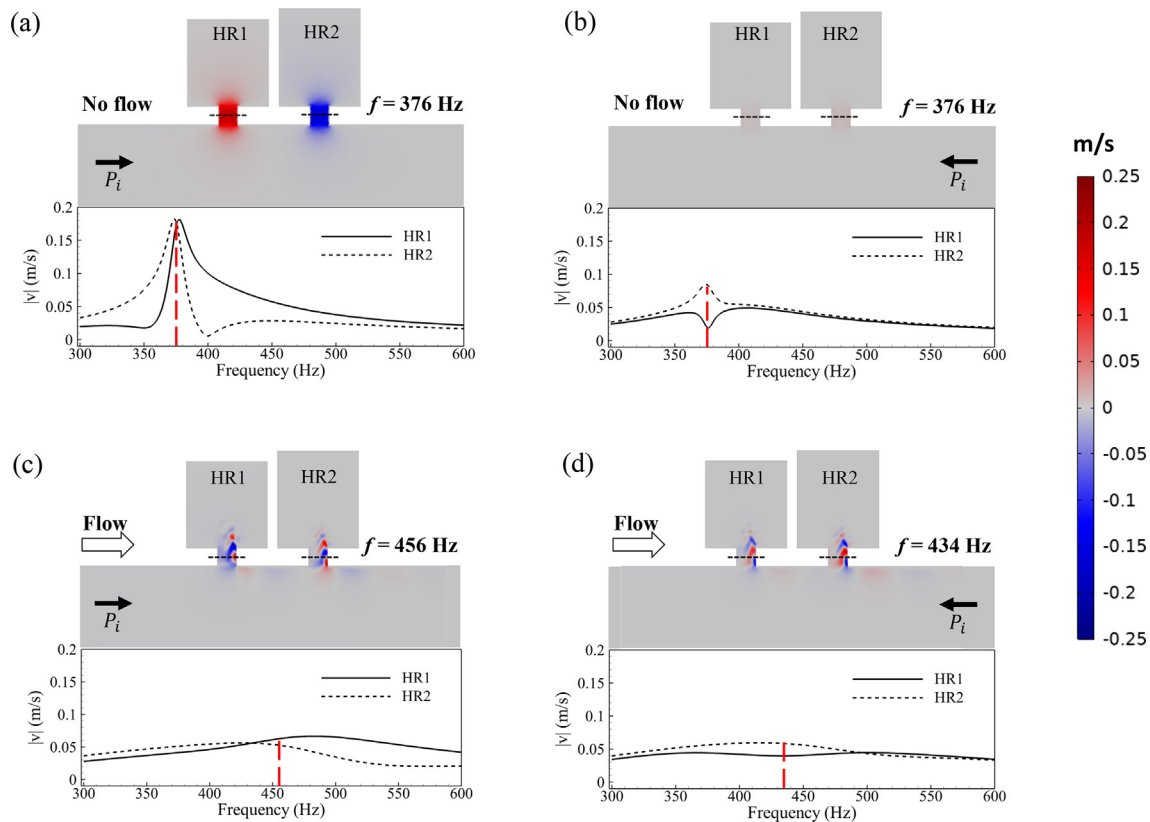


Fig. 11. Contours and spectrums of the sound perturbation velocity, the spectrum below is for the mean sound perturbation velocity magnitude of the black dash lines in the contours above, SPL = 110 dB. (a) Without flow, left incidence sound (Case 1), (b) without flow, right incidence sound (Case 2), (c) with flow, $M = 0.1$, left incidence sound (Case 3), (d) with flow, $M = 0.1$, right incidence sound (Case 4).

greater than its own geometric length, that is, the end correction in the impedance theory. However, when the flow is introduced (see Fig. 11(c)), not only the fluid at the neck end, but also the fluid inside the neck is reduced, which can be regarded as being blown into the main flow duct by the airflow. The reduction of the sound mass at the neck causes the local acoustic reactance to decrease, therefore, which shifts the resonant frequency to a high value.

5. Conclusions

Both numerical simulation and experimental measurements are conducted in this study. A three-dimension model of the double Helmholtz resonators (HRs) is established in the commercial finite element software, COMSOL Multiphysics 5.5. The sound transmission loss (STL) and sound scattering parameters ($R/T/A$) are predicted by solving the linearized Navier–Stokes (LNS) equations. A grazing flow impedance tube is fabricated to measure the acoustic characteristics of the coupled system. The two-microphone method considering the background flow is employed to derive the sound reflection coefficient. The sound attenuation performance is also evaluated based on the measured data. The experimental results agree with the numerical results very well. It is expected that the STL curve has two peaks for the no-flow cases, due to the coupled double HRs. However, there emerges only one peak of STL when adding the grazing flow in the duct. The corresponding peak values decline and the resonance frequency shifts to higher values. The system has a slightly better sound suppression performance when the sound wave and the grazing flow propagate in opposite directions than that in the same direction. Besides, the shear flow weakens the influence of sound incidence direction on acoustic performance, which invalidates the asym-

metric quasi-perfect absorptive/reflective feature of the system. The flow field contours show that the grazing flow impinges on the necks of HRs and causes an appreciable vortex in the necks and nearby the boundary layer. The sound absorption mechanism of the system is no longer just through thermal viscous damping, but considering the vortex sound interaction. In summary, the grazing flow damages the noise suppression performance of the coupled Helmholtz resonators. Under the shear flow, the sound incidence direction influences the resonant frequency and the peak value of sound transmission loss. Due to this inconsistency, the incoming direction of sound waves must be considered in the muffler design of the intake and exhaust system.

CRediT authorship contribution statement

Mingyang Zheng: Methodology, Software, Validation, Formal analysis, Writing - original draft. **Chao Chen:** Methodology, Writing - review & editing. **Xiaodong Li:** Conceptualization, Funding acquisition, Supervision, Writing - review & editing.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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